PROFESSOR APPROVAL SUBMISSION

Group Project Problem Statements

Submission Document for Course Approval in Simulation for Applied AI

COURSE
Simulation for Applied AI

INSTITUTION
Miami Dade College

GROUP MEMBERS
Jorge Mendoza, Jordan Gonzalez, Illia Shybanov, Ricardo Molina

REQUIREMENT
Mandatory 2 Problem Statements (Canvas Inbox)

> **Professor Directive:** Students MUST submit at least TWO problem statements for approval from the professor within ONE week after the assignment is available. The professor will decide whether a problem meets the requirements of the group project AND which problem the group should use as their project.

Problem Statement Options for Review

Option 1: Autonomous Mobile Robot (AMR) Fleet Logistics & Dynamic Recharging in the MDC AI & Robotics Lab

PREFERRED PROPOSAL

Real-World Setting	**Miami Dade College AI & Robotics Lab** (Classroom facility with shared hardware stations, component bins, and student project benches).
Problem Statement	During peak laboratory and capstone hours, students experience prolonged wait times (averaging **9.8 minutes** per request) waiting for microcontrollers, sensors, and electronic parts from the central inventory shelf (Station A), causing workbench bottlenecks at Soldering and QA Testing benches (Stations B & C). Concurrently, Autonomous Mobile Robots (AMRs) deployed to automate delivery suffer from reactive battery management (recharging only when depleted below 20%). This causes multiple AMRs to dock simultaneously at Station D, leaving zero active delivery units on the floor during peak class rushes and causing severe service interruptions.
Stakeholders	MDC Robotics/AI Students, Lab Instructors, Faculty, and Laboratory Technicians.
Multi-Method Approach (AnyLogic)	• **Discrete-Event Simulation (DES):** Models the arrival of student work orders (Poisson process), inventory queuing at Station A, and processing delays at workbench stations using AnyLogic's Process Modeling Library. • **Agent-Based Modeling (ABM):** Models individual AMRs (Inspector Alpha, Logistics Beta, Courier Gamma) with autonomous kinematics, spatial navigation on the MDC floorplan, collision avoidance, and battery discharge statecharts.

Reinforcement Learning (RL) Component	Compares **Tabular/Deep Q-Learning** against **Proximal Policy Optimization (PPO)** to solve the supervisory dispatch problem: determining whether an available AMR should service waiting station queues or execute an opportunistic "top-off" charge at Station D during momentary demand lulls.
3D Animation & Dashboard	Includes native AnyLogic 3D animation of the classroom and an interactive 5-tab **Streamlit Web Dashboard** with parameter sliders, real-time KPIs, and policy comparison graphs.
Target Outcomes	Cut average student wait times by >60%, completely eliminate simultaneous battery depletion stoppages, and identify the optimal fleet size (AMRs vs. charging pads).

Option 2: Smart Campus Autonomous Courier Rover Logistics for Miami Dade College Facilities

ALTERNATIVE PROPOSAL

Real-World Setting	**Miami Dade College Campus Grounds** (Inter-building transport between academic halls, administration, and campus health/science labs).
Problem Statement	Routine distribution of internal examinations, campus mail, specialized equipment, and medical/lab supplies between campus buildings relies heavily on manual courier staff, leading to delivery delays and high operational costs. Deploying small autonomous ground rovers introduces scheduling bottlenecks: rovers encounter pedestrian traffic congestion during class dismissal surges, experience battery depletion between distant campus facilities, and cause delivery delays if parcel dispatching is not synchronized with campus schedule cycles.
Stakeholders	MDC Campus Administration, Academic Department Staff, Campus Security, and Pedestrian Students.
Multi-Method Approach (AnyLogic)	• **Discrete-Event Simulation (DES):** Models intra-campus delivery request queues, priority sorting of urgent materials, and loading/unloading delays at building hubs. • **Agent-Based Modeling (ABM):** Models autonomous courier rovers navigating outdoor walkways, pedestrian avoidance zones, weather delays, and charging hub docking logic.
Reinforcement Learning (RL) Component	Compares **Q-Learning** vs. **PPO** to optimize multi-parcel batching, dynamic route selection around crowded walkways, and energy-efficient charging schedules.
3D Animation & Dashboard	Features AnyLogic 3D spatial animation of campus transit pathways alongside a Streamlit dashboard with fleet dispatch controls and route efficiency metrics.

Target Outcomes	Maximize on-time delivery percentages (>95%), minimize battery depletion downtime, and determine the optimal rover fleet size vs. charging depot footprint.

Canvas Submission Format Ready to Send

Below is the formatted text ready to copy-paste directly into the Canvas Inbox:

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Subject: Group Project Problem Statements for Approval - Team Jorge Mendoza, J

Dear Professor,

In accordance with the group assignment guidelines, our team has prepared two

OPTION 1 (Preferred): Autonomous Mobile Robot (AMR) Logistics & Dynamic Rechar
- Setting: Miami Dade College AI & Robotics Lab
- Problem: Student wait times for lab parts/equipment (~9.8 min average) and A
- Modeling: DES for work order queues at Stations A/B/C + ABM for AMR navigati
- RL: Comparing Q-Learning vs. PPO for supervisory dispatching and opportunist
- Desired Outcome: Reduce student wait times by >60% and determine the optimal

OPTION 2 (Alternative): Smart Autonomous Courier Rover Logistics for MDC Campu
- Setting: Miami Dade College Campus Grounds (inter-building delivery)
- Problem: Delays in transferring departmental supplies, lab samples, and mail
- Modeling: DES for parcel dispatch queues + ABM for courier rover agents trav
- RL: Comparing Q-Learning vs. PPO for parcel batching, walkway congestion avo
- Desired Outcome: Maximize on-time inter-building delivery and optimize fleet

Please let us know which problem statement you approve for our team to proceed

Thank you,
Jorge A Mendoza, Jordan Gonzalez, Illia Shybanov, Ricardo Molina Gonzalez
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